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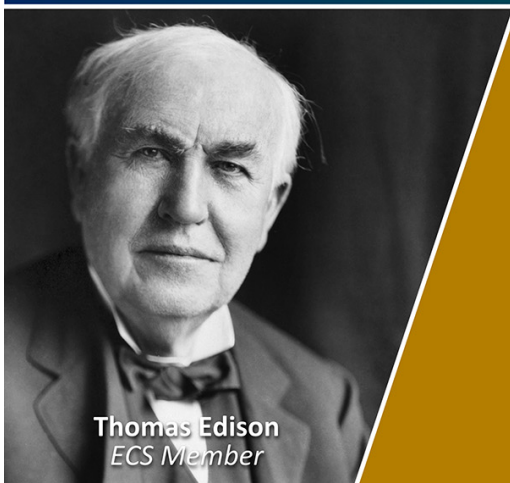
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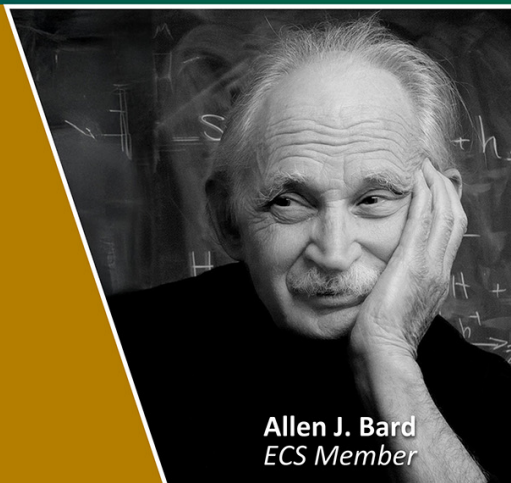


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Design and performance analysis of a flexible-hinged piezoelectric driving dispenser

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Abstract

Piezo-driven dispensers are precision dispensing devices for trace amounts of fluid and are widely used in the microelectronics packaging field. However, the preloading mechanism of piezo-driven dispensers easily loosens in high-frequency operation, and the design accuracy of the transmission ratio of the flexible displacement transmission mechanism (DTM) requires improvement. To address the aforementioned problems, we designed a self-locking preloading mechanism with an adjustable preloading amount for piezo-driven flexible transmission dispensers and investigated the method of designing the transmission ratio considering the non-expected directional deformation (parasitic deformation). First, the principle of the preloading mechanism was analyzed, and a method for adjusting the droplet diameter based on the preloading amount was proposed. Second, an asymmetric flexible-hinge compliance matrix calculation method was proposed, a transmission ratio model of the DTM was established, and the influence law of the structural parameters on the transmission ratio was comparatively analyzed using ANSYS software. Based on the output displacement requirement, transmission ratios were designed and structural parameters were determined. Finally, based on the piezoelectric coupling simulation, a piezo-driven dispenser was manufactured, and a test platform was built to conduct the relevant tests. The simulation and test results demonstrated that the maximum relative errors of the transmission-ratio calculated by the proposed transmission-ratio model were 1.54% and 3.6%, respectively, compared to the simulation and test results, confirming that the model was correct. Single-factor tests confirmed that a preloading mechanism can fix and preload the piezostack and that the droplet diameters can be adjusted with the preloading amounts. The operating frequency was up to 800 Hz with a diameter of 0.30 mm and a consistency of 4.32%, which meet the requirements of dispensing efficiency in the microelectronics packaging field. This study has practical significance for enhancing the transmission ratio design accuracy and dispensing performance of dispensers.

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Keywords: piezo-driven dispenser, self-locking preloading mechanism, compliance matrix method, transmission ratio, dispensing performance

1. Introduction

Fluid dispensing technology is widely used in cutting-edge fields, such as microelectronic packaging [1], inkjet printing [2, 3], and additive manufacturing [4, 5]. In the microelectronic packaging industry, fluid-dispensing technology plays a key role in chip encapsulation and fixation. Based on different driving methods, piezoelectric-driven [6, 7], pneumatic-driven [8, 9], and electromagnetic-driven [5, 10] dispensers are available. Among these, the piezo-driven type is widely used owing to its high frequency, fast response, and small volume of dispensing droplets. In addition, its operating frequency should be more than 100 Hz, and the needle stroke should meet the requirements of 0–300 μm to achieve better performance [11, 12].

In the field of piezo-driven dispensers for the microelectronic packaging, the piezostack actuation displacement is insufficient to satisfy the dispensing requirement [13]; therefore, a transmission mechanism is generally used to amplify it. The displacement transmission mechanisms (DTMs) of piezo-driven dispensers are mainly categorized into rigid [7, 14, 15] and flexible [16–20] mechanisms. A rigid DTM consists of a motion vice and connected rigid rods. The transmission of motion is realized through the movement of the rigid rods. Sun *et al* [7], Zhao *et al* [14], and Wang *et al* [21] designed a rigid transmission piezo-driven dispenser based on different amplification principles. They had a maximum operating frequency of up to 500 Hz, a radius of the droplets that remained within a range of 0.409–0.443 mm, and a droplet diameter consistency of <8%. The moving parts of a rigid DTM mutually wear and have gaps between them. Conversely, a flexible DTM uses the elastic deformation of the material to realize the transmission of motion [22]. It has the characteristics of fewer parts, no gaps between the moving parts, and no mutual wear; therefore, it is more widely used [23]. Bu *et al* [17] designed a triangular amplified flexible transmission dispenser, and its transmission ratio model was established according to the geometric relationship of the structure. Hu *et al* [18] designed a composite bridge amplified flexible transmission dispenser, and although the transmission ratio model was established by the pseudo-rigid-body method, its correctness was not verified. Zheng *et al* [19] designed a diamond-shaped amplified flexible transmission dispenser; its transmission ratio calculated according to the geometric relationship of the structure was compared with numerical simulation; the relative error between the calculation and numerical simulation was $\sim 21.9\%$. They had a maximum operating frequency of up to 400 Hz, a minimum droplet diameter of 0.525 mm, and a droplet diameter consistency of <5.62%.

Which is better than that of the rigid transmission piezo-driven dispenser. However, when analyzing the transmission

ratio, the influence of non-expected directional deformation (parasitic deformation) of the flexible hinge is ignored; therefore, the transmission ratio calculation error is large [24] and is not conducive to the structural design of the high-precision and miniaturized dispenser; it may even lead to an output displacement that does not meet the dispensing requirements. In addition, both rigid and flexible piezo-driven dispensers must fix and preload the piezostack to maintain a stable piezostack output force and displacement. Zhao *et al* [14], Sohn *et al* [16], Huang *et al* [25], Zhou *et al* [26], and Wang *et al* [27] used a bolt preloading mechanism to fix and preload the piezostack; the preloading direction of the bolts was parallel to the direction of the output force (displacement) of the piezostack; this could easily cause the preloading mechanism to loosen during the high-frequency vibration of the piezostack, resulting in poor consistency of the dispenser droplet or even failure to dispensing normally.

To address these two problems, a flexible transmission piezo-driven dispenser with self-locking and adjustable preloading mechanism is proposed in this paper. The transmission ratio that considers parasitic deformation was investigated in this study. First, a self-locking preloading mechanism with an adjustable preloading amount was designed; the working principle of the dispenser was analyzed; and a method for adjusting the diameter of the droplets by the preloading amount was proposed. Subsequently, based on the compliance matrix method, a transmission ratio calculation model of the DTM with an asymmetric flexible hinge was proposed; the influence of the structural parameters on the transmission ratio was analyzed by comparing with the numerical simulation model of ANSYS; consequently, the structural parameters of the transmission mechanism were determined. Finally, a dispenser was manufactured; a test platform was built to verify the accuracy of the proposed displacement transmission ratio model; and the dispensing performance of the dispenser was evaluated under different preloading amounts, operating frequencies, and supply pressures. The remainder of this paper is organized as follows: the materials and methods of this study are introduced in Chapter 2; the experimental results and discussion are provided in Chapter 3, and the conclusions of the materials and methods are stated in Chapter 4. The acronyms for this paper are listed in table 1.

2. Materials and methods

2.1. Structural design

Figure 1 illustrates a three-dimensional (3D) structure diagram that includes the preloading mechanism, piezostack, dispenser body, DTM, nozzle body, needle, and nozzle. The preloading

Table 1. List of acronyms.

Acronyms	Definition
PZT	piezoelectric stack
DTM	displacement transmission mechanism

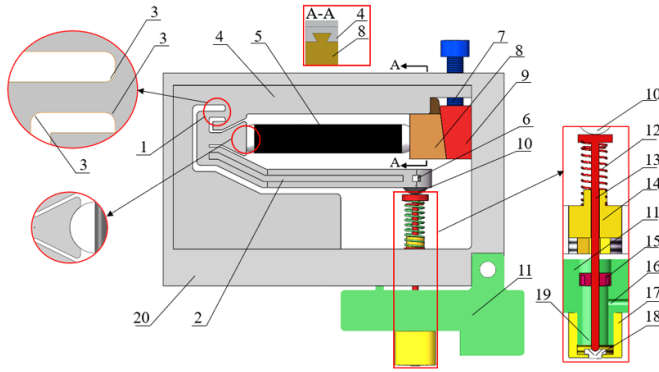


Figure 1. 3D design structure of piezo-driven dispenser: 1-flexible hinge, 2-L-type DTM, 3-transition chamfer, 4-dispenser body I, 5-piezostack, 6-test hole, 7-preloading bolt (M4 × 0.5), 8-preloading block B, 9-preloading block A, 10-needle I, 11-nozzle body, 12-return spring, 13-needle II, 14-guiding bolt, 15-oil sealing, 16-adhesive flow channel, 17-nozzle nut, 18-nozzle, 19-cavity, 20-dispenser body II.

mechanism consists of a preloading bolt and preloading blocks A and B. The preloading block B was matched with the dispenser body I through the dovetail groove; thus, the preloading block B exhibited only horizontal freedom of movement. The piezostack was fixed between the DTM and preloading mechanism; with both of its ends featuring spherical shapes to avoid the tangential force, and the preloading force of the piezostack was adjusted by the preloading bolt. The DTM is a part of dispenser body I. It utilizes the triangular amplification principle of the L-type mechanism to maximize the use of horizontal space; thus, it is compact. The flexible part of the DTM adopted a flexible leaf hinge, and the connection part with the rigid body adopted a chamfer to reduce stress. The end ball head of the DTM was needle I, which constituted a double-needle structure with needle II. Needle II maintained contact with needle I through the return spring, and the pre-pressure of the return spring was adjusted through the guiding bolt. The nozzle body of the dispenser has an adhesive flow channel inside; thus, the adhesive flowed into the cavity under supply pressure. During long-term use, the nozzle, needle II, and oil sealing are vulnerable to wear and failure. To facilitate replacement, the nozzle-body assembly was designed to be detachable and fixed to the dispenser body using screws. The oil seal had an interference fit with the needle II and nozzle body to avoid adhesive leakage.

Figure 2 illustrates a schematic of the preloading mechanism. Its working principle is as follows: first, screwing the bolt pushes the preloading block A to move downward (① in figure 2). Subsequently, the preloading block B, under

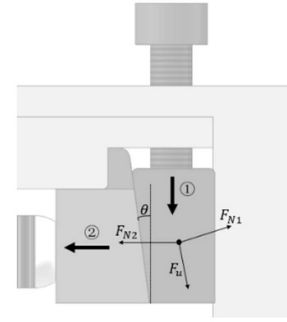


Figure 2. Schematic of preloading mechanism.

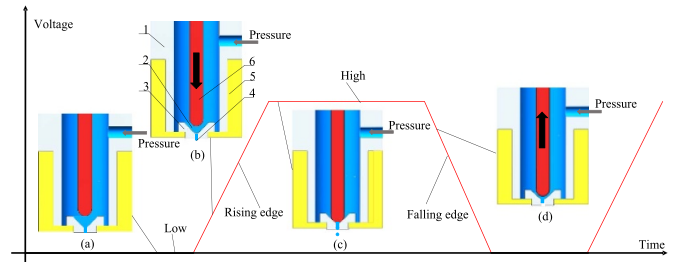


Figure 3. Schematic of the dispensing process of a piezo-driven dispenser. 1-nozzle body, 2-nozzle chamber, 3-nozzle, 4-adhesive, 5-nozzle nut, 6-needle II. (a) Needle II lifting up. (b) Needle II lowering. (c) Needle II hits nozzle and droplet forms. (d) Needle II rising.

the motion of preloading block A, moves to the left and squeezes the piezostack to form the preload force (② in figure 2). Conversely, since the working vibration direction of piezostack is orthogonal to the preloading direction of the preloading bolt, the influence of the working vibration of piezostack on the preloading bolt is reduced. Since the inclination angle of the contact surfaces between preloading blocks A and B is smaller than the friction angle, the preloading block can self-lock under the vibration force of the piezostack. Thus, it prevents the preloading mechanism from loosening. Simultaneously, the preload force increases with the increasing preloading amount. When the preload force exceeds a certain threshold, the piezostack pushes the flexible hinge rotation; the needle I of DTM and needle II then move downward; thus, the distance between the needle II and nozzle changes; then the droplet diameter changes due to the change in the motion stroke of the needle II. Thus, the phenomenon of frequent screwing of the nozzle nut caused the sealing to deteriorate and led to adhesive leakage can be avoided.

Figure 3 illustrates a schematic of the dispensing process under a square-wave input signal. The working process is as follows: when the input signal is low, piezostack has no output displacement, the needle II keeps lifting up, as shown in figure 3(a), and the adhesive flows out from the nozzle under the supply pressure. When the input signal increases to a high level, the piezostack quickly elongates, and the flexible hinge deforms with clockwise microrotation under the action of the

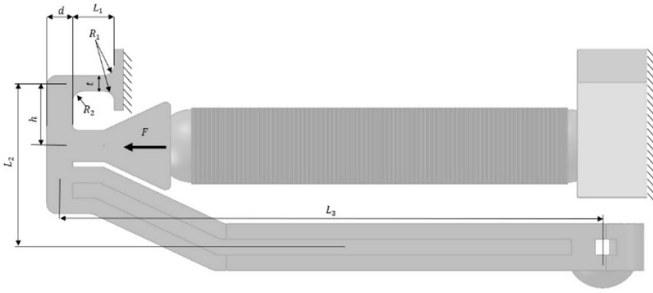


Figure 4. Schematic of the DTM of the piezo-driven dispenser.

DTM. This pushes the needle II to overcome the spring force (glucose viscous force), and then downward movement; more adhesive is extruded from the nozzle, as shown in figure 3(b). When needle II hits the nozzle, the adhesive separates and forms a droplet when its kinetic energy is sufficiently high, as shown in figure 3(c). When the input signal was high, the nozzle was blocked and no adhesive flowed out as needle II was always in contact with the nozzle. When the input signal gradually reduces, the piezostack shrinks rapidly; subsequently, the DTM is reset under the restoring force of the flexible hinge; and the needle II is gradually restored to the lifting position under the restoring force of the return spring, as shown in figure 3(d). Simultaneously, vacuum is formed in the nozzle chamber, and the adhesive quickly fills the chamber under the supply pressure, and then waits for the next cycle.

2.2. Modeling and analysis

When designing a flexible DTM under the influence of parasitic deformation, a large error is observed between the theoretical and actual calculations, which influences the accuracy of the structural design. Therefore, in this study, a transmission ratio model of the DTM with an asymmetric flexible hinge was established based on the compliance matrix method; and the transmission ratio and structural parameters were designed according to the output displacement requirements. Simultaneously, to ensure the safe operation of the DTM, the strength of the flexible hinge was calibrated.

2.2.1. Transmission ratio calculation of DTM. (1)

Compliance Matrix: figure 4 illustrates a diagram of the piezo-driven DTM of the dispenser. R_1 and R_2 are the transition chamfer radii, L_1 is the flexible hinge length, d is the minimum thickness of the rigid body part, t is the minimum thickness of the flexible hinge, h is the distance between the center of the flexible hinge and equivalent action position of the piezostack output force, w is the width of the DTM, and L_3 is the length of the DTM output cantilever. When analyzing the transmission ratio, only the compliance matrix of the flexible hinge must be solved as it undergoes elastic deformation.

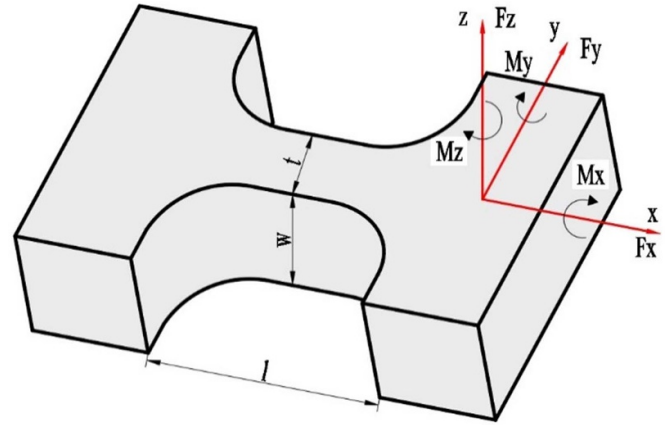


Figure 5. Schematic of standard round corner type flexible hinge under force.

Figure 5 illustrates a schematic of a standard rounded flexible hinge under force and moment. The deformation of the flexible hinge is regarded as the result of the superposition of the deformations of many segments, and the compliance matrix is calculated according to the Euler–Bernoulli beam deformation theory [28].

For a planar flexible-hinge mechanism, the deformation in the plane is expressed as

$$\varepsilon = C * F, \quad (1)$$

where ε denotes the deformation matrix of the flexible hinge, F is the matrix of forces and moments on the flexible hinge, C is the compliance matrix of the flexible hinge.

Deformation matrix ε is

$$\varepsilon = \begin{bmatrix} \varepsilon_x & \varepsilon_y & \alpha_z \end{bmatrix}^T, \quad (2)$$

where ε_x is the linear deformation of the flexible hinge in the x -axis direction (mm); ε_y is the linear deformation of the flexible hinge in the y -axis direction (mm); and α_z is the angular deformation of the flexible hinge around the z -axis (rad).

Force F on the flexible hinge is expressed as

$$F = \begin{bmatrix} F_x & F_y & M_z \end{bmatrix}^T, \quad (3)$$

where F_x is the force on the flexible hinge in the x -axis direction (N); F_y is the force on the flexible hinge in the y -axis direction (N); M_z is the moment on the flexible hinge around the z -axis (N mm).

By substituting equations (2) and (3) into equation (1), the compliance matrix C can be expressed as

$$C = \begin{bmatrix} C_{11} & 0 & 0 \\ 0 & C_{22} & C_{23} \\ 0 & C_{32} & C_{33} \end{bmatrix} = \begin{bmatrix} \frac{\varepsilon_x}{F_x} & 0 & 0 \\ 0 & \frac{\varepsilon_y}{F_y} & \frac{\varepsilon_z}{M_z} \\ 0 & \frac{\alpha_z}{F_y} & \frac{\alpha_z}{M_z} \end{bmatrix}. \quad (4)$$

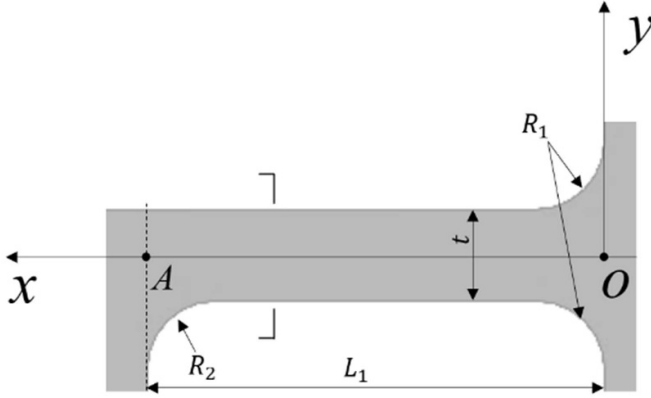


Figure 6. Asymmetric flexible hinge and coordinate system.

Figure 6 shows a schematic of the flexible hinge (chamfered asymmetric flexible hinge) and coordinate system used in this study. The OA section is a variable cross-section; the thickness of the hinge in the y -direction varies with the x -axis; and the thickness $h(x)$ is given by

$$h(x) = \begin{cases} t + 2R_1 - 2\sqrt{R_1^2 - (R_1 - x)^2}, & x \in (0, R_1) \\ t, & x \in (R_1, L_1 - R_2) \\ t + R_2 - \sqrt{R_2^2 - (R_2 - x)^2}, & x \in (L_1 - R_2, L_1) \end{cases} \quad (5)$$

where t is the minimum thickness of the flexible hinge (mm); R_1 and R_2 are the radii of the transition chamfer of the flexible hinge (mm).

The angular deformation α_z along the z axis, which is the most important deformation parameter, is generated by the forces M_z and F in figure 5. The angular deformation α'_z under the action of the M_z is given by

$$\begin{aligned} \alpha'_z &= \int_0^{L_1} \frac{M_z}{EI_z(x)} dx \\ &= \int_0^{R_1} \frac{12M_z}{Ewh^3(x)} dx + \int_{R_1}^{L_1-R_2} \frac{12M_z}{Ewt^3} dx \\ &\quad + \int_{L_1-R_2}^{L_1} \frac{12M_z}{Ewh^3(x)} dx, \end{aligned} \quad (6)$$

where E is the modulus of elasticity of the material (Pa); w is the width of the rectangular cross-section of the flexible hinge (mm); and $I_z(x)$ is the moment of inertia of the microelement cross-section to the z -axis (mm^4).

Associating equation (4) with equation (6), the compliance C_{33} corresponding to the moment M_z is expressed as

$$C_{33} = \frac{12}{Ew} \left(\int_0^{R_1} \frac{1}{(t + 2R_1 - 2\sqrt{R_1^2 - (R_1 - x)^2})^3} dx + \frac{L_1 - R_1 - R_2}{t^3} + \int_{L_1-R_2}^{L_1} \frac{1}{(t + R_2 - \sqrt{R_2^2 - (L_1 - R_2 - x)^2})^3} dx \right) \quad (7)$$

The microelement moment of the flexible hinge corresponding to the force F_y is $F_y(L_1 - x)$. Therefore, the angular deformation α''_z of the flexible hinge corresponding to F_y can be expressed as

$$\begin{aligned} \alpha''_z &= \int_0^{L_1} \frac{F_y(L_1 - x)}{EI_z(x)} dx \\ &= \int_0^{R_1} \frac{12F_y(L_1 - x)}{Ewh^3(x)} dx + \int_{R_1}^{L_1-R_2} \frac{12F_y(L_1 - x)}{Ewt^3} dx \\ &\quad + \int_{L_1-R_2}^{L_1} \frac{12F_y(L_1 - x)}{Ewh^3(x)} dx. \end{aligned} \quad (8)$$

Associating equation (4) with equation (8), the compliance C_{32} corresponding to the force F_y can be expressed as

$$C_{32} = \frac{12}{Ew} \left(\int_0^{R_1} \frac{L_1 - x}{(t + 2R_1 - 2\sqrt{R_1^2 - (R_1 - x)^2})^3} dx + \int_{R_1}^{L_1-R_2} \frac{L_1 - x}{t^3} dx + \int_{L_1-R_2}^{L_1} \frac{L_1 - x}{(t + R_2 - \sqrt{R_2^2 - (L_1 - R_2 - x)^2})^3} dx \right). \quad (9)$$

The deformation ε_x is produced by the force F_x and can be expressed as

$$\begin{aligned} \varepsilon_x &= \int_0^{L_1} \frac{F_x}{Ewh(x)} dx \\ &= \int_0^{R_1} \frac{F_x}{Ewh(x)} dx + \frac{F_x(L_1 - R_1 - R_2)}{Ewt} \\ &\quad + \int_{L_1-R_2}^{L_1} \frac{F_x}{Ewh(x)} dx. \end{aligned} \quad (10)$$

Associating equation (4) with equation (10), the compliance C_{11} corresponding to the force F_x can be expressed as

$$C_{11} = \frac{1}{Ew} \left(\int_0^{R_1} \frac{1}{t + 2R_1 - 2\sqrt{R_1^2 - (R_1 - x)^2}} dx + \frac{(L_1 - R_1 - R_2)}{t} + \int_{L_1 - R_2}^{L_1} \frac{1}{t + R_2 - \sqrt{R_2^2 - (L_1 - R_2 - x)^2}} dx \right), \quad (11)$$

The deformation ε_y is produced by the moment M_z and force F_y . Assuming that the compliance matrix is symmetric, the compliance C_{23} corresponding to the moment M_z is expressed as

$$C_{23} = \frac{\varepsilon_y'}{M_z} = C_{32} = \frac{\alpha_z''}{F_y}, \quad (12)$$

where ε_y' is the linear deformation in the y-axis direction corresponding to the moment M_z (mm).

The linear deformation ε_y'' corresponding to the force F_y mainly consists of the bending deformation ε_y^b and shear deformation ε_y^s , and can be expressed as

$$\begin{aligned} \varepsilon_y'' &= \varepsilon_y^b + \varepsilon_y^s \\ &= \int_0^{L_1} \tan \alpha_z'' dx + \int_0^{L_1} \frac{kF_y}{Gwh(x)} dx \\ &= \frac{F_y}{Ew} \int_0^{L_1} \int_0^x \frac{(L_1 - y)}{h^3(y)} dy dx + \frac{kF_y}{Gw} \int_0^{L_1} \frac{1}{h(x)} dx, \end{aligned} \quad (13)$$

where k is the material shear coefficient and G is the shear modulus of the material (Pa).

Associating equation (4) with equation (13), the y-axis line deformation compliance C_{22} corresponding to the force F_y is expressed as

$$C_{22} = \frac{1}{Ew} \int_0^{L_1} \int_0^x \frac{(L_1 - y)}{h^3(y)} dy dx + \frac{k}{Gw} \int_0^{L_1} \frac{1}{h(x)} dx \quad (14)$$

(2) Transmission ratio model: figure 7 illustrates a simplified analytical model of the DTM. $OA'B'D'E'$ represents the DTM model before deformation, and $OABDE$ represents the DTM model after deformation, where $C(C')$ is the intersection point of $AB(A'B')$ and $DE(D'E')$, F_x is the output force of the piezostack, $B(B')$ is the equivalent action position of the output force of the piezostack on the DTM, and $E(E')$ is the output point of the DTM.

The input displacement of the DTM can be expressed as an $\Delta x_{B'} = x_B - x_{B'}$ along the x-axis, and its output displacement can be expressed as an $\Delta y_{E'} = y_E - y_{E'}$ along the y-axis. Therefore, the ratio of the DTM is given as

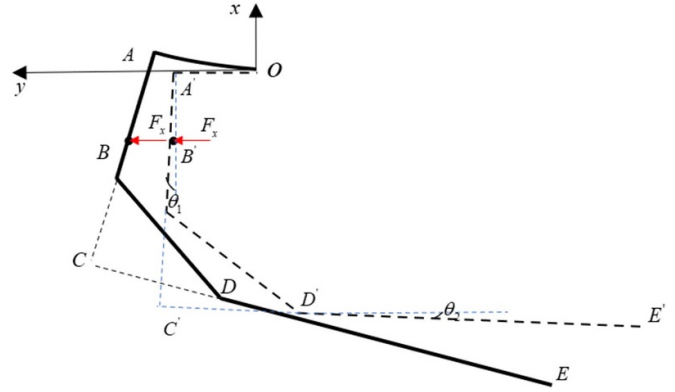


Figure 7. Simplified analytical model of the DTM.

$$R_{amp} = \frac{\Delta y_{E'}}{\Delta x_{B'}}. \quad (15)$$

Assuming that the coordinates of the points before deformation are given as

$$\begin{cases} A'(l_{OA'}, 0) \\ B'(l_{OA'} + l_{A'B'} \sin \theta_1, -l_{A'B'} \cos \theta_1) \\ C'(l_{OA'} + l_{A'C'} \sin \theta_1, -l_{A'C'} \cos \theta_1) \\ E'(l_{OA'} + l_{A'C'} \sin \theta_1 - l_{C'E'} \cos \theta_2, -l_{A'C'} \cos \theta_1 - l_{C'E'} \sin \theta_2) \end{cases} \quad (16)$$

where $l_{OA'}$, $l_{A'B'}$, $l_{A'C'}$ and $l_{C'E'}$ are the lengths of $O'A'$, $A'B'$, $A'C'$ and $C'D'$ sections, respectively; θ_1 is the angle between $A'B'$ and y-axis; and θ_2 is the angle between $D'E'$ and x-axis. After the output force of piezostack is applied, the deformation of the flexible hinge at point A is $\varepsilon_A = [\varepsilon_x \ \varepsilon_y \ \alpha_z]^T$; except for $O'A'$ section, which is a flexible part, the rest of the DTM is considered as a rigid body; then, the rigid body along the z-axis rotational deformation is α_z ; the coordinates of each point can be expressed as equation (17) after the deformation:

$$\begin{cases} A(l_{OA'} + \varepsilon_x, \varepsilon_y) \\ B(l_{OA'} + \varepsilon_x + l_{A'B'} \sin(\theta_1 + \alpha_z), \varepsilon_y - l_{A'B'} \cos(\theta_1 + \alpha_z)) \\ C(l_{OA'} + \varepsilon_x + l_{A'C'} \sin(\theta_1 + \alpha_z), \varepsilon_y - l_{A'C'} \cos(\theta_1 + \alpha_z)) \\ E(l_{OA'} + \varepsilon_x + l_{A'C'} \sin(\theta_1 + \alpha_z) - l_{C'E'} \cos(\theta_2 + \alpha_z), \varepsilon_y - l_{A'C'} \cos(\theta_1 + \alpha_z) - l_{C'E'} \sin(\theta_2 + \alpha_z)) \end{cases} \quad (17)$$

Associating equation (16) with equation (17), the coordinate relationship before and after the deformation can be expressed as

$$\begin{cases} \Delta x_B = \varepsilon_x + l_{A'B'} \sin(\theta_1 + \alpha_z) - l_{A'B'} \sin \theta_1 \\ \Delta y_E = \varepsilon_y - l_{A'C'} \cos(\theta_1 + \alpha_z) - l_{C'E'} \sin(\theta_2 + \alpha_z) \\ \quad + l_{A'C'} \cos \theta_1 + l_{C'E'} \sin \theta_2 \end{cases} \quad (18)$$

The piezostack motion displacement is at the micrometer level, and the flexible hinge produces only micro deformation, which satisfies the small deformation assumption combined with the equivalent infinitesimal relation ($\alpha_z \sim \sin \alpha_z$).

Substituting equation (18) into equation (15) yields the transmission ratios expressed by equation (19):

$$R_{\text{model}} = \left| \frac{\varepsilon_y - l_{A'C'} * \alpha_z * \sin \theta_1 - l_{C'E'} * \alpha_z * \cos \theta_2}{\varepsilon_x + l_{A'B'} * \alpha_z * \cos \theta_1} \right|. \quad (19)$$

Based on equations (6)–(14) and (19), the transmission ratio only has relation to the structural parameters of the DTM and has no relation to the preloading influences of the preloading amount.

2.2.2. Flexible hinge strength calibration. Neglecting the influence of the shear force in the y -direction of the flexible hinge, the main forces on the flexible hinge are the combined tensile and bending loads, and the maximum stress can be expressed as [29]:

$$\sigma_{\text{max}} = \sigma_s + \sigma_{b\text{max}}, \quad (20)$$

where σ_s is the tensile stress (Pa); $\sigma_{b\text{max}}$ is the maximum bending stress (Pa).

σ_s and $\sigma_{b\text{max}}$ are expressed as equation (21):

$$\begin{cases} \sigma_s = K_s \frac{F_x}{wt} \\ \sigma_{b\text{max}} = K_b \frac{6(M_z + F_y L_1)}{wt^2} \end{cases} \quad (21)$$

where K_s is the theoretical stress concentration factor for the axial load; K_b is the theoretical stress concentration factor for the bending load.

The stiffness and flexibility matrices are inverse of each other; therefore, the stiffness matrix $\mathbf{K}$ can be expressed as equation (22) using the compliance matrix $\mathbf{C}$.

$$\mathbf{K} = \begin{bmatrix} K_{11} & 0 & 0 \\ 0 & K_{22} & K_{23} \\ 0 & K_{32} & K_{33} \end{bmatrix} = \mathbf{C}^{-1}. \quad (22)$$

Based on the relationship between the flexible-hinge deformation, load, and stiffness factor, the maximum stress σ_{max} is given by

$$\begin{aligned} \sigma_{\text{max}} &= \frac{K_s K_{11} \varepsilon_x}{wt} + \frac{6K_b}{wt^2} [(L_1 K_{22} + K_{32}) \varepsilon_y + (L_1 K_{23} + K_{33}) \alpha_z]. \end{aligned} \quad (23)$$

To ensure the safe operation of the DTM, the maximum combined stress of the flexible hinge should be maintained within the permissible stress of the material.

$$\sigma_{\text{max}} < \frac{\sigma_y}{n_a}, \quad (24)$$

where σ_y is the yield stress (Pa); and n_a is the safety factor, generally 1.5–2.5.

Table 2. Main structural parameters of DTM.

Parameter	Value	Parameter	Value
h (AB)	6 mm	w	14 mm
L_1 (OA)	4 mm	R_1	0.5 mm
L_3 (CE)	52 mm	R_2	0.5 mm
d	3 mm	θ_1	0°
t	1 mm	θ_2	0°

2.3. Numerical simulation analysis

2.3.1. Influence simulation of structural parameters.

ANSYS software was used to establish the numerical simulation model for the DTM; the basic structural parameters and material properties of the DTM are listed in tables 2 and 3, and the parameter symbols corresponding to the symbols in figures 4, 6, and 7 are included in table 2.

To enhance the simulation efficiency, only dispenser body I in figure 1 was numerically simulated. The face in contact with the dispenser bodies I and II was set as fixed constraint, the mesh division type was tetrahedron, and the mesh cell size was set as 1×10^{-3} m. In the simulation, the force was applied at the position where the piezostack acts on the DTM; the input displacement of the DTM was set to $\sim 38 \mu\text{m}$; and the transmission ratio with the output displacement of the simulation was calculated. The basic structural parameters are listed in table 2. The sizes of L_3 , h , L_1 , t , d , and R_1 (R_2) parameters in the simulation model were changed to analyze the influence law of the structural parameters on the transmission ratio.

2.3.2. Displacement and strength simulation of DTM. The output displacement of the piezostack with respect to the output force under driving voltage satisfies equation (25) [30]:

$$\begin{cases} \Delta L = nd_{33}U - \frac{F_x}{k_{pzt}} \\ \Delta L_0 = nd_{33}U \end{cases} \quad (25)$$

where ΔL_0 and ΔL are the no-load and actual output displacements of the piezostack, respectively (μm); d_{33} is the piezoelectric constant (C/N); n is the number of piezostack layers; U is the piezostack driving voltage (V); F_x is the piezostack axial output force (N); k_{pzt} is the piezostack axial equivalent stiffness ($\text{N } \mu\text{m}^{-1}$).

With the action of the output force of the piezostack, the flexible hinge of the DTM has elastic deformation, forming a variable force load on the piezostack. Based on the small deformation assumption, the variable force load can be equated to a spring load. According to the equilibrium relationship of forces:

$$\begin{cases} \Delta L_s = \Delta L_0 \left(\frac{k_{pzt}}{k_{pzt} + k_s} \right) \\ \Delta L_l = \Delta L_0 \left(1 - \frac{k_{pzt}}{k_{pzt} + k_s} \right) \end{cases} \quad (26)$$

Table 3. Material properties of dispenser assemblies.

Material Parameter	Steel	Aluminum alloy	Carbide	Ceramic
Density (kg m^{-3})	7850	2770	7800	3900
Poisson ratio	0.3	0.3	0.25	0.36
Young's modulus (Pa)	2.010^{11}	7.2×10^{10}	5.7×10^{11}	5.5×10^{11}
Bulk modulus (Pa)	1.67×10^{11}	6.0×10^{10}	3.8×10^{11}	6.54×10^{11}
Shear modulus (Pa)	7.69×10^{10}	2.77×10^{10}	2.28×10^{10}	2.02×10^{11}

Table 4. The technical parameters of adopted piezostack.

Dimension	Nominal Displacement	Blocking Force
$7 \times 7 \times 36 \text{ mm}$	$38 \mu\text{m}$	1960 N
Capacitance	d33	Voltage
$7.2 \mu\text{f}$	$7.1'10\text{-}10 \text{ C/N}$	120 V

where ΔL_s and ΔL_l are the actual displacement and output displacement loss of the piezostack, respectively (μm); k_s is the equivalent stiffness of the variable force load ($\text{N } \mu\text{m}^{-1}$), which is mainly determined by the rotational compliance C_{33} of the flexible hinge. According to equation (26), the actual output displacement of the piezostack with variable force load is less than the no-load displacement, and the output displacement loss ΔL_l is very small because of $k_s \ll k_{pzt}$.

To dispense a certain viscosity range of the adhesive, the maximum stroke of the needle must satisfy equation (27):

$$\Delta x = \Delta x_B = \Delta L_s * R_{\text{model}} \geq [\Delta x_t], \quad (27)$$

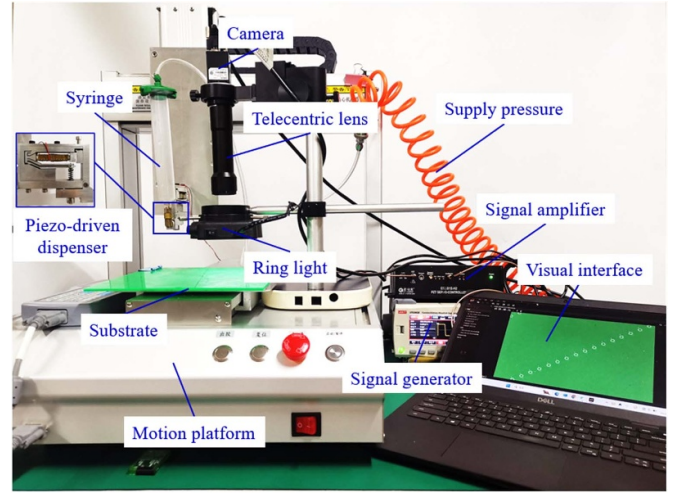
where $[\Delta x_t]$ is the stroke threshold of needle (μm).

According to the needle stroke requirement and strength calibration analysis, the thickness of the flexible hinge t was set to 1.5 mm; and the remaining structural and material parameters are listed in tables 2 and 3, respectively. The piezostack type was a DJF73601 (COREMORROW), and its parameters are listed in table 4.

In this study, a piezoelectric coupling simulation of the DTM and piezostack was performed using ANSYS software. The contact type of the piezostack with the DTM and preloading block was set to no separation, and the mesh division type was tetrahedral. The mesh cell size of the flexible hinge area was $5 \times 10^{-4} \text{ m}$, and the other areas were set as default; the grid transition was set to slow. Piezostack was applied with 120 V driving voltage. The maximum stress of the flexible hinge and the displacements (input and output) of the DTM were analyzed under the above boundary conditions.

2.4. Experiment

2.4.1. Test platforms and instruments. In this study, to verify the correctness of the ratio model and operating performance of the dispenser, a trial of piezo-driven dispenser was created. A test platform as shown in figure 8 was built. The length and diameter of the needle II were 40.5 mm and $\phi 1.5 \text{ mm}$, respectively, and the diameter of the nozzle was $\phi 0.15 \text{ mm}$.

**Figure 8.** Piezo-driven test platform.

An air compressor (EWS 24–680 W type, ELUAN) was used to provide the supply pressure. A signal generator (UTG-2000B type, UNI-TREND Technology) was used to generate a square-wave input signal, which was then linearly amplified using a signal amplifier (E52.B1S-H type, COREMORROW) to drive the piezostack cycle movement. A three-axis moving platform drove the substrate to deposit the adhesive droplets. The diameter of the adhesive droplets was measured using an industrial camera (MV-CS200-10GC type, HVISION, 20 megapixels, 5472×3648 resolution) with a telecentric lens (MVL-KT-15-110 type, HVISION, $110 \pm 1 \text{ mm}$ working distance). The output displacement of the DTM was measured by a laser micrometer (CDX-L15A type, OPTEX, $\pm 1 \text{ mm}$ measuring range, $\pm 0.05\%$ linear accuracy).

2.4.2. Validation of transmission ratio model. As illustrated in figure 9, the output displacement test platform was built under the condition that the structural parameters were determined. The driving voltage was taken as the test factor; the output displacement of the DTM (and needle) was taken as the test index; and the relative error was used to evaluate the correctness of the transmission ratio model. As the body of the dispenser blocks the laser micrometer beam, the output displacement of the DTM is equivalently expressed by measuring the displacement of the vertical movement of the test rod fixed to the end of the DTM. The operating frequency was set to 5 Hz; the driving voltage was 45, 60, 75, 90, 105, and 120 V in

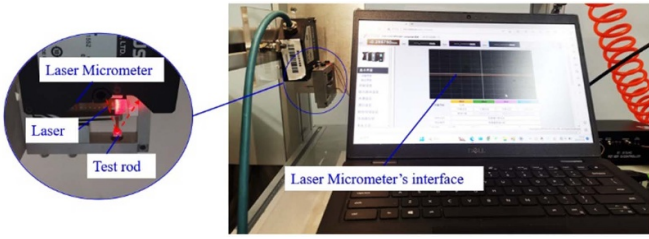


Figure 9. Output displacement test platform.

order. The output displacement was tested; each set of measurements was repeated three times; and after averaging, the actual transmission ratio was calculated as

$$R_{\text{real}} = \frac{\Delta x_{\text{DTM}}}{\Delta y_{\text{PZT}}}, \quad (28)$$

where Δx_{DTM} and Δy_{PZT} are the output displacements of the DTM and piezostack (also input displacement of DTM), respectively. As it cannot be measured directly, the no-load output displacement of the piezostack stated by the manufacturer was used (μm).

2.4.3. Single-factor tests for dispensing performance.

Based on the dispensing characteristics of the adhesive, a simplified model [25] of a single dispensing volume of the piezo-driven dispenser is given as

$$V_{\text{out}} = \frac{\pi \Delta p R_{\text{nozzle}}^4}{8uL_{\text{nozzle}}} t_{\text{on}}, \quad (29)$$

where u is the dynamic viscosity of the adhesive (Pa s); L_{nozzle} is the length of the nozzle hole (mm); R_{nozzle} is the diameter of the nozzle hole (mm); Δp is the pressure difference between the inner and outer surfaces of the nozzle hole (Pa); t_{on} is the open time of the nozzle (s). According to equation (29), if the parameters of the nozzle structure and the viscosity of the adhesive are constant, Δp and t_{on} affect the single-dispensing volume of the piezo-driven dispenser. Therefore, in this study, we changed Δp through the supply pressure and t_{on} through the operating frequency and preloading amount of the preloading mechanism to perform the related single-factor test. The test parameters are listed in table 5. The basic operating conditions were set to: a driving signal with a high level: 90 V, duty cycle: 50%, and rising time: 120 μs . When the distance between the nozzle and substrate was fixed, the deposited droplets were hemispherical crown-shaped [9]; therefore, the droplet diameter and its consistency were selected as the test indices in this study. A mixture of glycerin and alcohol with a mixing ratio of 1.5:1 was used as the experimental object. During the preloading amount adjustment, the preloading bolt was first adjusted to the piezostack just to be fixed (preloading amount of 0°), and then the preloading amount was adjusted to 90° , 180° , 270° , and 360° in order.

The testing process was as follows: First, we set the control parameters of the motion platform (three-axis motion speed, distance between the dispenser and substrate, dispensing time,

Table 5. Parameters of single-factor test.

Test group	Test factor		
	Preloading amount (Degrees)	Operating frequency (Hz)	Supply pressure (bar)
1	$0^\circ, 90^\circ, 180^\circ, 270^\circ, 360^\circ$	300	4
2	90°	100, 200, 300, 400, 500	4
3	90°	300	1.5, 2, 3, 4, 5, 6, 7

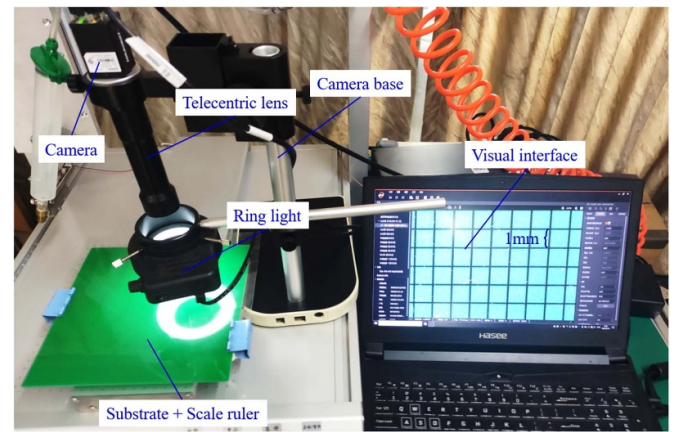


Figure 10. An image of the industrial camera calibration.

etc.) and dispenser (operating frequency, preloading amount, supply pressure, etc.). Then, we dispensed over 200 droplets at a time for these control parameters and selected 20 droplets at random. Finally, we measured each diameter of these droplets, averaged them as the droplet diameter, and calculated the diameter consistency and its standard deviation for that combination of factor levels. The consistency of the droplet diameter was evaluated based on the relative error between the maximum (minimum) and the mean value of the droplet diameter. This testing process was done one by one, and the distance between the dispenser and substrate was fixed throughout the whole test. Other parameters of the motion platform were adjusted as the operating frequency changed. All the control parameters that affect droplet morphology were kept constant by one combination of factors. Thus, consistency calculations are scientific.

To ensure the accuracy of the measurement of the droplet diameter, the industrial camera was calibrated. Figure 10 shows an image of the industrial camera calibration. The distance between the lens and substrate was adjusted to 11 cm; the pixel micron size of the camera was calibrated by a scale paper; the calibrated camera field of view was $8.75 \times 5.84 \text{ mm}^2$, and the pixel micron size was $1.6 \times 1.6 \mu\text{m}^2$. Therefore, the adhesive droplet diameter D can be expressed as

$$D = \frac{d_1 + d_2}{2} \times 1.6 \times 10^{-3}, \quad (30)$$

where d_1 is the droplet diameter in the vertical direction (mm); d_2 is the droplet diameter in the horizontal direction (mm); D is the average droplet diameter (mm).

3. Result and discussion

3.1. Static simulation analysis

3.1.1. Influence simulation of structural parameters on transmission ratio. Figure 11 illustrates the influence of different structural parameters on the transmission ratio. According to figure 11(a), L_3 was approximately proportional to the transmission ratio, significantly influenced the size of the dispenser structure, and sufficiently large to significantly increase the overall size. According to figure 11(b), the transmission ratio decreased with the increasing distance h , which approximated a quadratic relationship. However, h cannot be significantly reduced owing to the limitation of the piezostack size. Figures 11(c), (d) and (f) show the influence of the flexible hinge length L_1 , minimum thickness t , and radii of the transition chamfer (R_1 and R_2) on the transmission ratio, which have lesser influence on the transmission ratio than L_3 and h . According to figure 11(c), the transmission ratio decreased with an increase in L_1 . The increase in L_1 results in an increase in the axial and upward tangential parasitic deformation of the flexible hinge and reduces the transmission ratio. According to figures 11(d) and (f), the parameters t , R_1 , and R_2 had less influence on the transmission ratio; however, they significantly influenced the stress of the flexible hinge. Therefore, they need to be considered to prevent the flexible hinges from working safely when too small.

Since the theoretical model only considers the all-directional deformation of the flexible hinge, the theoretical results have some differences from the simulation. According to equation (7), the compliance C_{33} (the main factor affecting the deformation of the flexible hinge) increased as L_1 increased, and then the required input force decreased with the same input displacements ($\sim 38 \mu\text{m}$). Thus, the deformation of the rigid body part decreased, and the relative error of the transmission ratio decreased, which had a maximum relative error of 1.54% when L_1 was 1 mm, as shown in figure 11(c). The compliance C_{33} also decreased as t increased; thus, the relative error of the transmission ratio increased, as shown in figure 11(d), with a maximum relative error of $\sim 0.5\%$. According to figure 11(e), the relative error of the transmission ratio decreased as d increased. With the action of the input force, the rigid body part will undergo a larger deformation if d was small. When the d was >2.5 mm, the theoretical and numerically calculated transmission ratios approximated each other, and the deformation of the non-flexible hinge part could be neglected.

In summary, the simulation and the theoretical model analyze the effect of structural parameters on the transmission ratio with the same trend, but some differences exist between them under the deformation influence of the non-flexible hinge part. The difference can be reduced with the distance L_1 , the thickness d of the non-flexible hinge

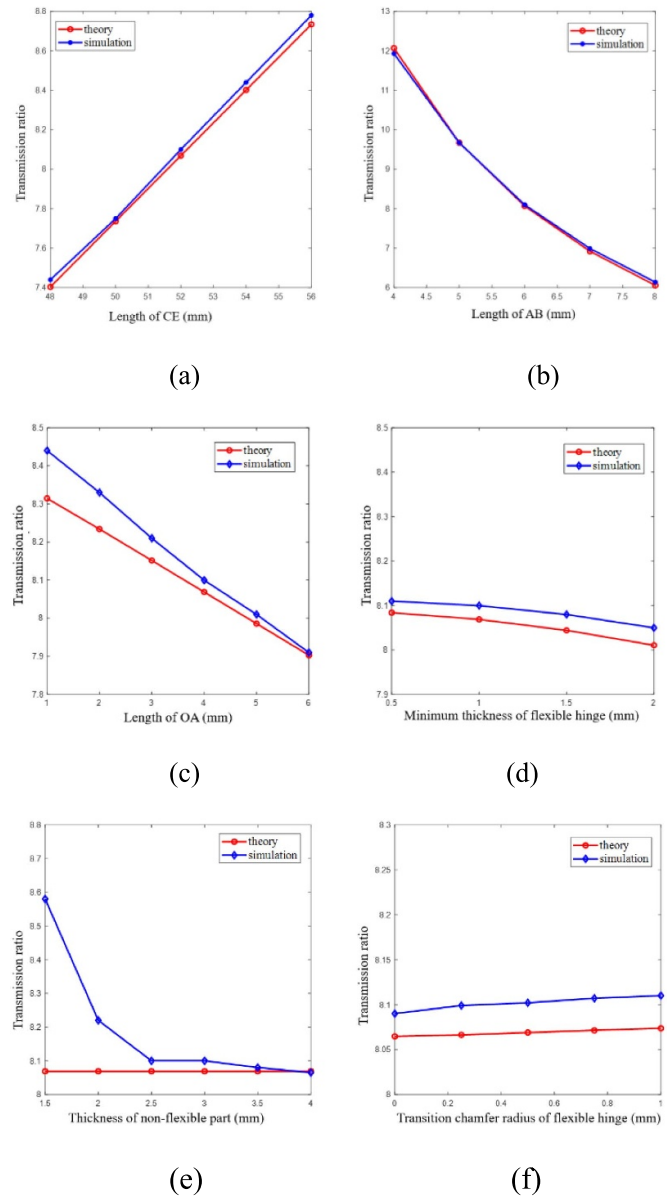


Figure 11. Influence curve of structural parameters on transmission ratio. (a) Influence curve of the cantilever length (L_3) of DTM on transmission ratio. (b) Influence curve of the distance between the equivalent position of the piezostack output force and the center of the flexible hinge (h) on the transmission ratio. (c) Curve of flexible hinge length (L_1) on transmission ratio. (d) Curve of flexible hinge thickness (t) on transmission ratio. (e) Curve of the non-flexible part thickness (d) on the transmission ratio. (f) Influence curve of transition chamfer (R_1 and R_2) on transmission ratio.

part increased, or the thickness t of the flexible hinge decreased. However, the size of DTM increased as the parameter d or L_1 increased, and the internal stress of the flexible hinge increased as the parameter t decreased. Thus, when the parasitic deformation of the flexible hinge was considered and the structural parameters were designed appropriately, the DTM can meet the output displacement requirement and be more compact, accurate, and durable in design.

3.1.2. Displacement and strength simulation of DTM.

Figure 12 shows a cloud diagram of the ANSYS simulations. Under the conditions of the structural parameters, mentioned in section 2.3.2, after a maximum driving voltage of 120 V was applied to the piezostack. According to figure 12(a), the maximum stress of the DTM was 83.81 MPa, which was made of aluminum alloy 6061 and has a permissible stress of 145 MPa. According to figure 12(b), the maximum stress of the piezostack was 15.67 MPa, which was made of ceramic and has a permissible internal stress of 50 MPa [31]. Thus, the maximum stress of the piezostack and designed DTM was less than the permissible stress, and a large margin ensured the safe operation of the key components. According to figure 12(c), the output displacement of piezostack (also the input displacement of DTM) was 36.74 μm . The relative error between the simulated output displacement of the piezostack subjected to the load of the DTM and the nominal no-load output displacement was 3.3%. This indicated that the error introduced by using the no-load output displacement of the piezostack to represent the actual displacement in equation (28), was small. The output displacement of DTM affects the ultimate performance of the dispenser, such as droplet diameter, adhesive viscosity, etc. When the control parameters of the dispenser are constant, the greater the output displacement of the DTM, the greater the needle stroke, the larger the droplet diameter, and the greater the viscosity of adhesive that can be dispensed. According to figure 12(d), the output displacement of DTM was $\sim 298.3 \mu\text{m}$, which basically met the requirements of piezo-driven dispenser in the field of microelectronics packaging for the output stroke of 0–300 μm [11, 12].

3.2. Experiment of transmission ratio and dispensing performance

3.2.1. Transmission ratio testing. The ideal transmission ratio derived from the geometrical relationship of the DTM is given as [17, 19]

$$R_{\text{ideal}} = \frac{L_3}{h}, \quad (31)$$

where h is the distance between the center of the flexible hinge and equivalent output force acting on the piezostack (mm), and L_3 is the output cantilever length of the DTM (mm).

The output displacements of the DTM, measured using the laser micrometer under different driving voltages, are listed in table 6.

For the DTM designed in this study, the actual transmission ratio under different driving voltages was 7.76–8.11, with an average value of 7.91, as shown in figure 13. When the parasitic deformation was neglected, the ratio calculated using equation (31) was $R_{\text{ideal}} = 8.67$, and the maximum relative error with respect to the actual value was 10.5%. After considering parasitic deformation, the ratio calculated using

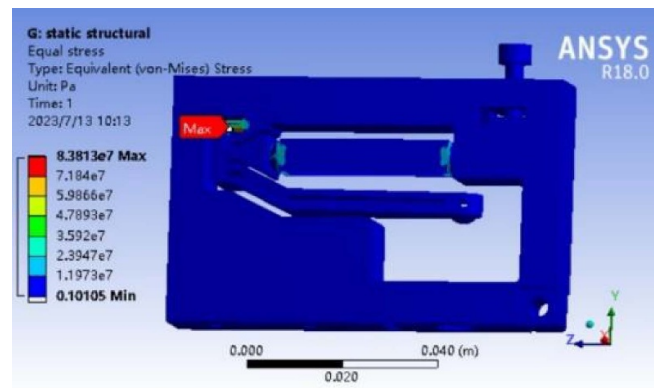
equation (19) was $R_{\text{model}} = 8.05$, and the maximum relative error with the actual value was 3.6%. Thus, consideration of parasitic deformation improved the accuracy of the ratio calculation by 6.9%. Thus, it verified the accuracy of the ratio calculation model in this study. In addition, the theoretically calculated ratio was slightly larger than the actual value, as the actual output displacement of the piezostack was slightly smaller than the no-load output displacement under the influence of the DTM load. In summary, the calculation of the transmission ratio using the structural geometric relationship can be used for the preliminary design of the DTM, whereas the effect of parasitic deformation must be considered for a precise and compact design. Consequently, when we neglect parasitic deformation in the flexible DTM design, the output displacement deviates from the design target more notably, which is more likely to affect the dispenser's ultimate performance or even fail to meet its design requirements.

3.2.2. Single-factor influence test on dispensing performance.

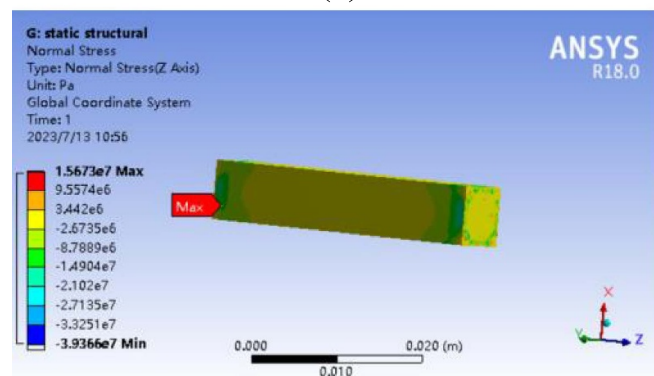
(1) Influence of preloading amount on dispensing performance: As shown in figure 14(a), the preloading amount significantly affected the droplet diameter. As the preloading amount was gradually increased from 0 to 1 circle, the diameter of the droplets decreased from 1.44 to 0.35 mm. According to the working principle of the preloading mechanism mentioned in section 2.1, as the stroke of the needle decreased with an increase in the preloading amount, the downward movement time of the needle shortened and the volume of the adhesive being thrust out of the nozzle by the needle reduced; thus, the diameter of the adhesive droplets reduced. However, when the preloading amount was significantly large, the small distance between the needle and nozzle resulted in an excessively short acceleration time and insufficient kinetic energy of the adhesive; thus, proper dispensing was not achieved. As shown in figure 14(b), the consistency and standard deviation of the droplet diameter under different preload amounts were within 3% and 2.5%, respectively. Meanwhile, under the same level of influencing factors, no loosening of the preloading mechanism occurred after multiple sets of tests. This indicated that the preloading mechanism could effectively fix and preload the piezostack.

In this study, the method of adjusting the distance between the needle II and the nozzle through the preloading amount is simple and reliable, which provides a new method for adjusting the distance between the needle and the nozzle and keeps the distance stable with the self-locking mechanism, and the phenomenon of frequent screwing of the nozzle nut that caused the sealing to deteriorate and led to adhesive leakage can be avoided. The self-locking mechanism is also suitable for the other high-frequency working scenarios of the piezostack that need to be fixed and preloaded.

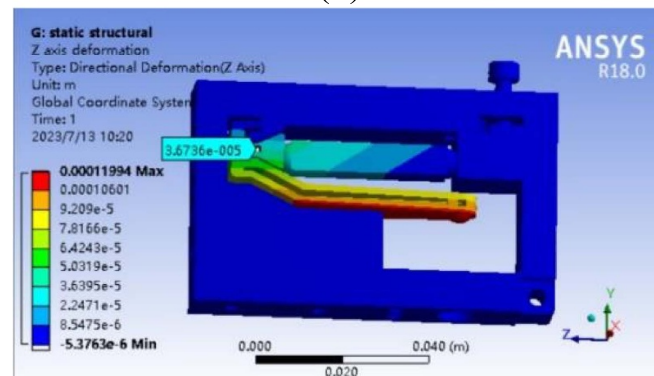
(2) Influence of operating frequency on dispensing performance: With the development of microelectronics



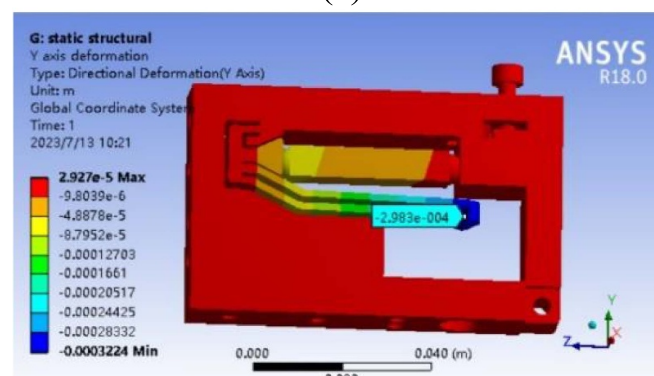
(a)



(b)



(c)

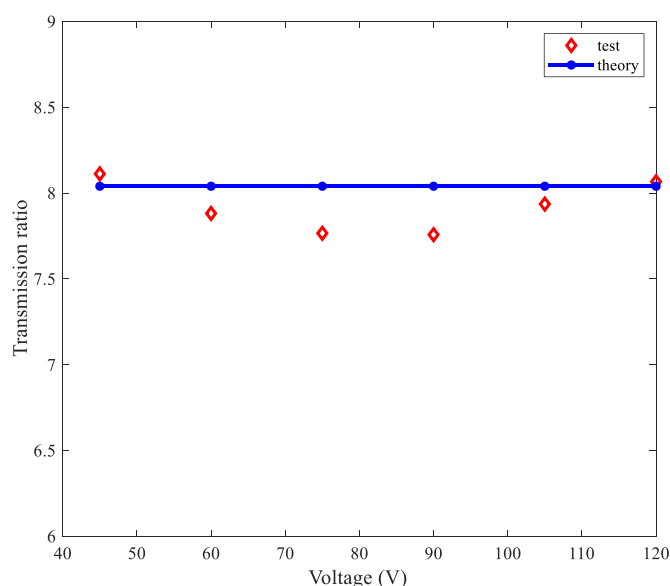


(d)

Figure 12. Cloud map of ANSYS numerical simulation. (a) Stress map of DTM. (b) Stress map of piezostack. (c) Input Displacement of the DTM. (d) Output displacement of the DTM.

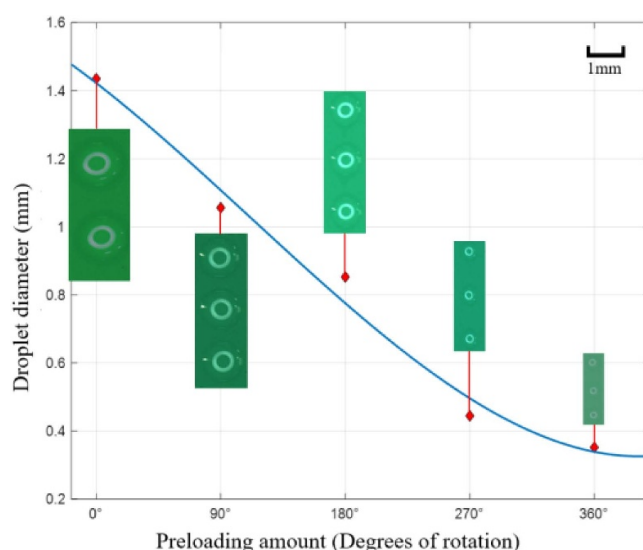
Table 6. Reference displacement of piezostack and output displacement of DTM under different driving voltages.

driving voltage (V)	45	60	75	90	105	120
Piezostack output displacement (μm)	14.67	19.85	24.85	29.26	33.20	36.88
DTM output displacement (μm)	119	156.5	193	227	263.5	297

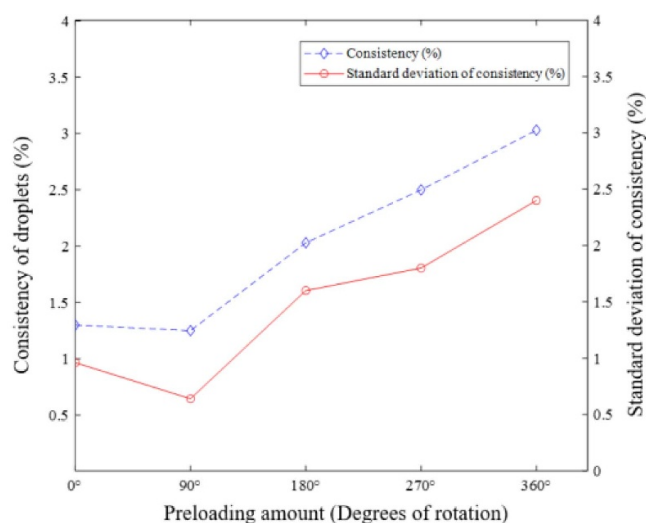
**Figure 13.** Distribution of actual values of transmission ratio under different driving voltage conditions.

technology, the required operating speed is getting faster and faster. Thus, operating frequency is a very important indicator. When the operating frequency was gradually increased from 100 to 500 Hz, the diameter of the droplet gradually reduced from 1.42 to 0.65 mm, as shown in figure 15(a). As the operating frequency increased, the nozzle opening time shortened; therefore, the adhesive dispensing volume under the supply pressure reduced, and the diameter of the adhesive decreased. For a combination of control parameters, the dispenser dispensing more than 200 droplets, as shown in figure 15(b). When the frequency exceeded 500 Hz, adhesive sticking occurred, as shown in figure 15(c). Since the piezostack is a capacitive load, its response time was insufficient when the operating frequency was significantly high; at this point, the needle did not have enough time to accelerate, and the actual movement stroke of the needle was considerably small. Therefore, the kinetic energy of the adhesive dispensing out of the nozzle could not overcome the viscous resistance between the adhesive and nozzle to form a droplet. This indicated that the piezo-driven dispenser was limited to a certain parameter condition in terms of the operating frequency.

To further evaluate the high-frequency performance of the piezo-driven dispenser, the rise time of driving signal was set to 40 μs , the preloading amount was adjusted, and the operating frequency of dispenser was up to 700–800 Hz, as shown in figure 16. At this time, the dispenser stably dispensed the adhesive, and the droplets exhibited a well-formed shape. This



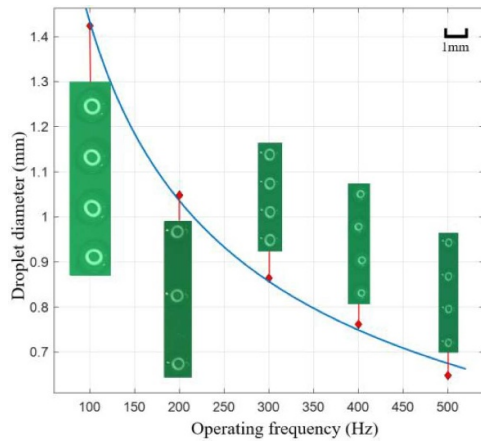
(a)



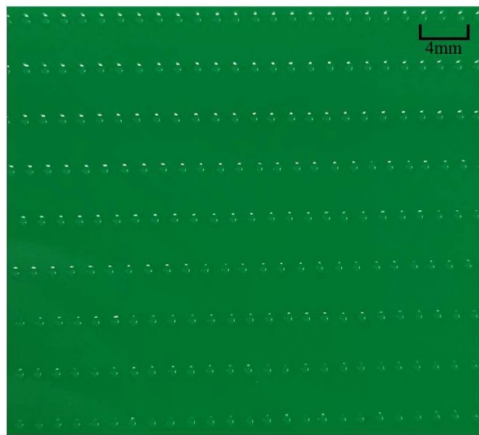
(b)

Figure 14. Influence of preloading amount of preloading mechanism on dispensing performance. (a) Influence curve of the preloading amount on the droplet diameter. (b) Distribution of the droplet diameter consistency and its standard deviation with different preloading amount.

shows that as the rise time of the driving signal decreased, the response time of the piezostack accelerated, and the needle could obtain a sufficient movement stroke and kinetic energy in a shorter time to realize normal dispensing.



(a)



(b)



(c)

Figure 15. Influence of operating frequency on dispensing performance. (a) Curve of the operating frequency influence on droplet diameter (b) overall image of adhesive droplets (300 Hz) (c) adhesive hanging phenomenon.

For the operating frequency of 800 Hz, the droplet diameter was 0.30 mm, the relative error of the droplet diameter consistency was $\sim 4.32\%$, and the standard deviation of consistency was $\sim 3.1\%$. This indicated that the droplet diameter consistency was good at high frequencies. As the operating frequency continued to increase, the signal amplifier began to overload; therefore, the output capacity of the driving power supply significantly influenced the maximum operating

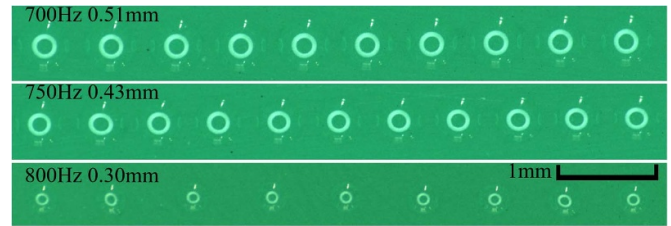
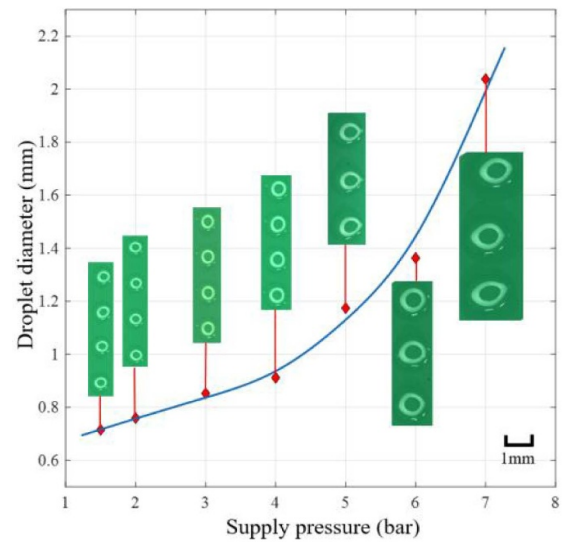
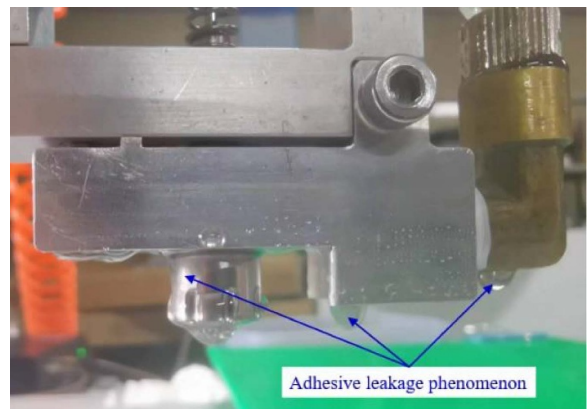


Figure 16. Image of an adhesive droplets at 700–800 Hz.



(a)



(b)

Figure 17. Influence of supply pressure on dispensing performance. (a) Influence curve of supply pressure on droplet diameter. (b) Adhesive leakage phenomenon.

frequency. In summary, the simultaneous adjustment of several control parameters enabled a higher dispenser operating frequency.

(3) Influence of supply pressure on dispensing performance: The diameter of the adhesive droplet increased gradually from 0.72 to 2.04 mm when the supply pressure increased from 1.5 to 7 bar, as shown in figure 17(a). On one hand, as

the supply pressure increased, more adhesive dispensed out of the nozzle when the needle was lifted; however, less adhesive overcame the supply pressure and receded back into the syringe when the needle moved downward; thus, the droplets that dispensed out of the nozzle were larger in diameter. At the same time, the downward movement resistance of the needle also increased as the supply pressure increased, which is more unfavorable to the high-frequency work of the dispenser. When the supply pressure exceeded 7 bar, the adhesive leaked and could not be properly dispensed, as shown in figure 17(b). When the supply pressure was <1.5 bar, an adhesive hanging phenomenon occurred as the supply pressure was insufficient to fill the vacuum in time in the nozzle hole and upper cavity caused by the needle lifting. In addition, when the needle moved downward and dispensed the adhesive, most of the adhesive receded back into the syringe owing to the insufficient supply pressure; thus, the volume of adhesive dispensing out of the nozzle was too small and hanging phenomenon occurred.

4. Conclusion

In this study, a flexible-transmission piezo-driven dispenser containing a self-locking preloading mechanism with an adjustable preloading amount was designed, and a displacement transmission ratio design method considering parasitic deformation was investigated. First, a beveled self-locking preloading mechanism is proposed to solve the problem, in which the mechanism tends to loosen during the high-frequency operation of the piezo-driven dispenser. The proposed preloading mechanism can adjust the diameter of the adhesive droplets. Second, a method for calculating the compliance matrix of an asymmetric flexible hinge was proposed and a ratio model of the DTM was obtained. Finally, a trial production prototype was tested based on theoretical calculations and simulation results. The simulation and test results showed that the maximum relative errors of the proposed ratio model with the simulation and experimental tests were 1.54% and 3.6%, respectively. This verified the correctness of the transmission ratio model and provides a solution for the transmission ratio design of the flexible DTM of the piezo-driven dispenser. The results of the single-factor test showed that the preloading mechanism could fix and preload the piezostack, the maximum operating frequency of the dispenser was up to 800 Hz, the droplet diameter was 0.30 mm, and the consistency of the droplet size was 4.32%, and the standard deviation of consistency was ~3.1%. This verified the feasibility of adjusting the droplet diameter (volume) by the preloading amount of the preloading mechanism, and the performance of the dispenser met the requirements of the industry. Due to the power limitation of the signal amplifier, it was not possible to test the higher operating frequencies of the dispenser. In the future, we will attempt to improve the transmission efficiency of a DTM and analyze its dynamic response performance.

Data availability statement

The data cannot be made publicly available upon publication because they are not available in a format that is sufficiently accessible or reusable by other researchers. The data that support the findings of this study are available upon reasonable request from the authors.

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Conflict of interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

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